

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #15

Modeling Concepts

Ceyda Oğuz
coguz@ku.edu.tr

Koç University

Agenda

- ❑ Announcement
- ❑ Models
- ❑ Decision Making Process
- ❑ Mathematical Models
- ❑ Mathematical Programming Models
 - Example

Announcement

- ❑ Midterm will be held on 25 April 2026, Saturday, between 11:45-14:45.
- ❑ Venue will be ENG Z50.
- ❑ You will be seated according to a plan.
- ❑ Only essential materials (pencil, eraser, calculator, ID and water) will be allowed. All belongings should be left at the podium of the classroom.
- ❑ No phones, watches or any digital equipment are allowed.

Announcement – cont'd

- ❑ It will be a closed-book and closed-notes exam.
- ❑ No cheat-sheet is necessary.
- ❑ You are not allowed to ask any questions during the exam.
- ❑ Duration of the exam will be around 120 minutes. It will not be permitted to leave and come back during the exam.

Announcement – cont'd

- ❑ Midterm will include all topics we covered till Lecture 15 – Modeling (today's lecture).
- ❑ There will be different types of questions:
 - Quantitative: methodology of IE.
 - Qualitative: concepts of IE.
 - Questions will be in the format of test questions: multiple choice, fill-in-the-blanks, true-false
- ❑ *You must show your work clearly and write legibly to receive full credit.*
- ❑ There are a set of study questions uploaded at LearnHub.

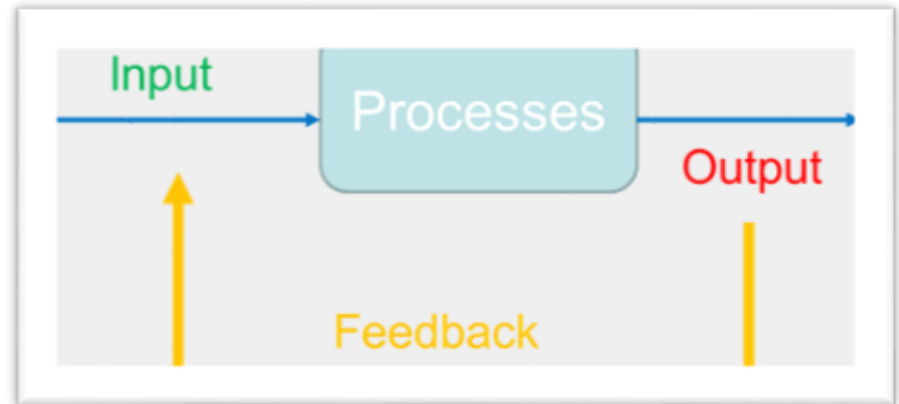
Where are we?

- ❑ We learnt how to design production facilities
 - Focusing on process
 - Capacity
 - Layout
 - Work design
- ❑ Next, you have to make decisions about your system
 - Production planning
 - Distributing your products

} Mathematical approach
(Operations Research)

What is next?

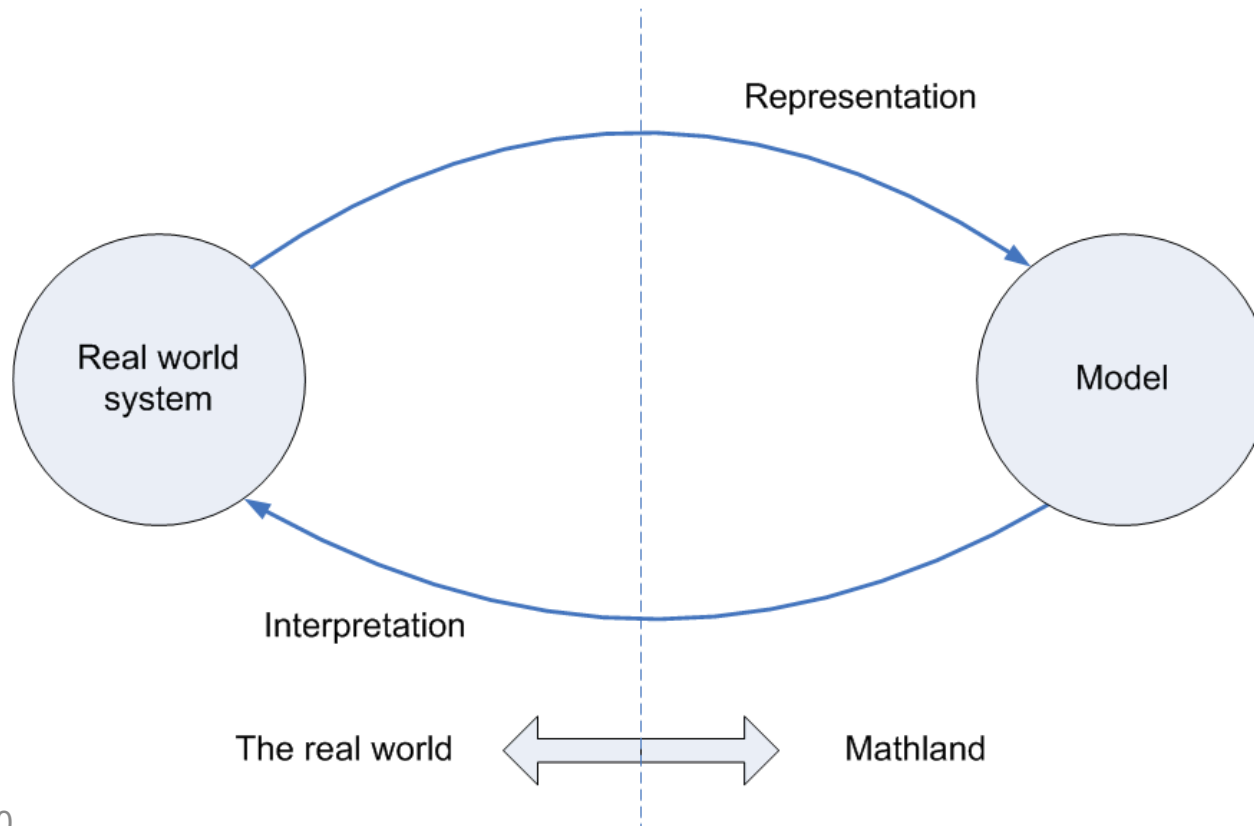
- ❑ We have a system



- ❑ A production system is anything that takes inputs and transforms them into outputs.
- ❑ To use mathematics, we need a model of the system.

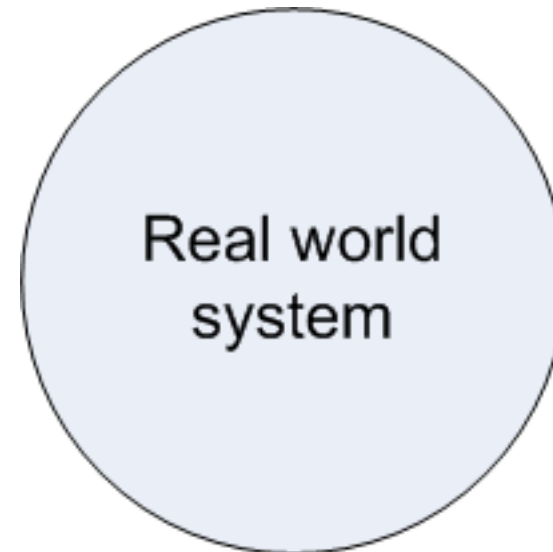
What is a model?

- Models are representations of real objects or systems
- Building a model helps understanding a system



Real world system

- Borders between the system to analyze and the environment:
 - What do we control?
 - What do we take as given?



Representation

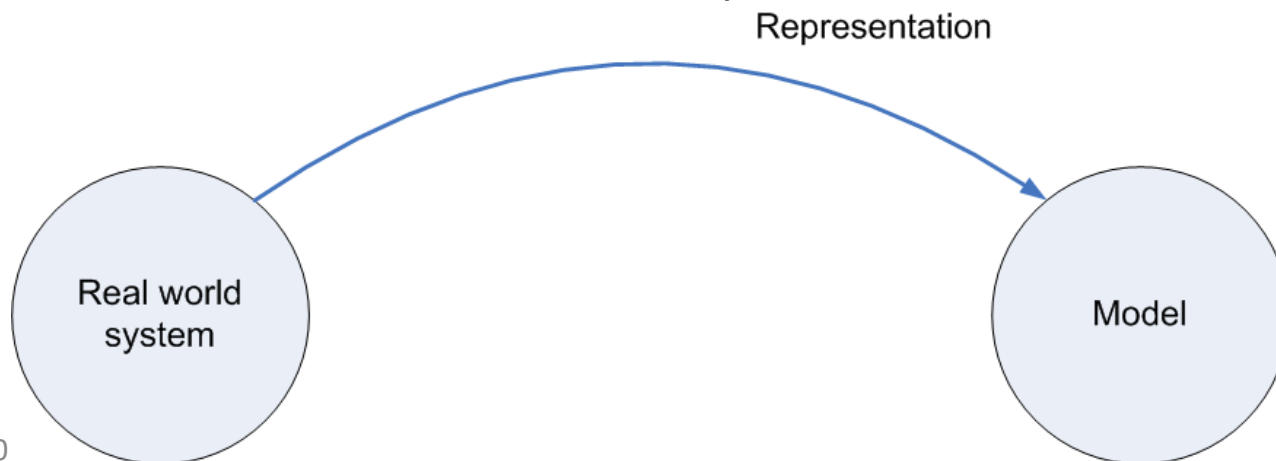
- Physical models
 - crash dummies



Models are representations of real objects or systems

- For IEs: mathematical models:
 - Relevant components of the system
 - Level of detail versus tractability of the model

Building a model helps understanding a system

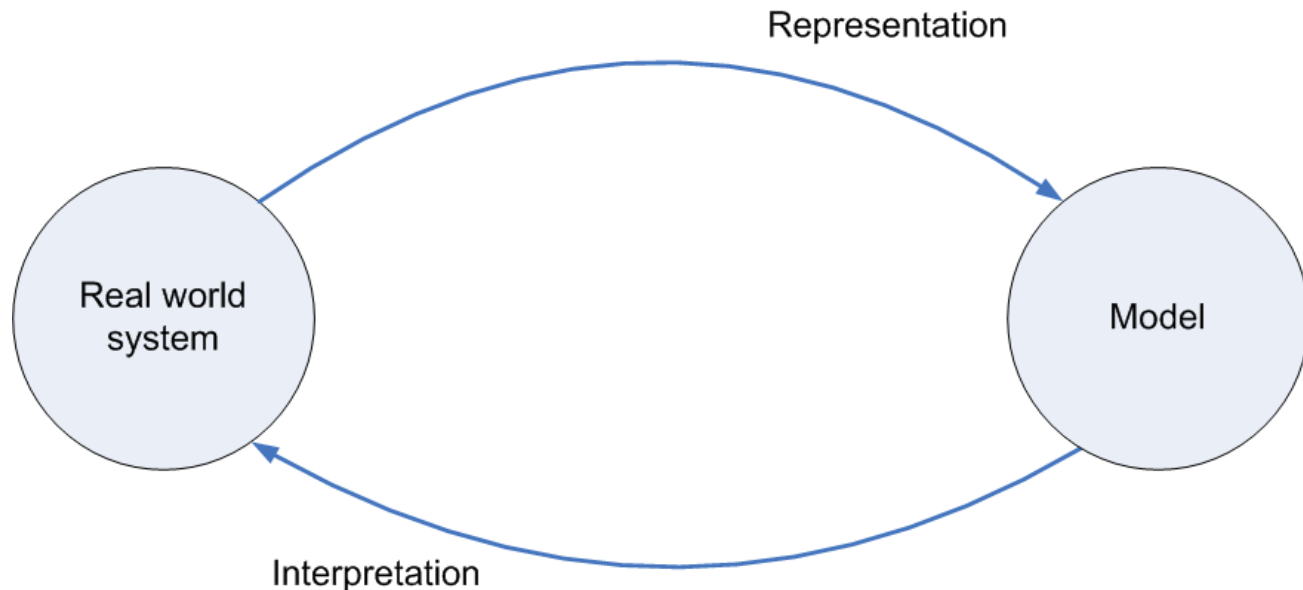


Why do we need models?

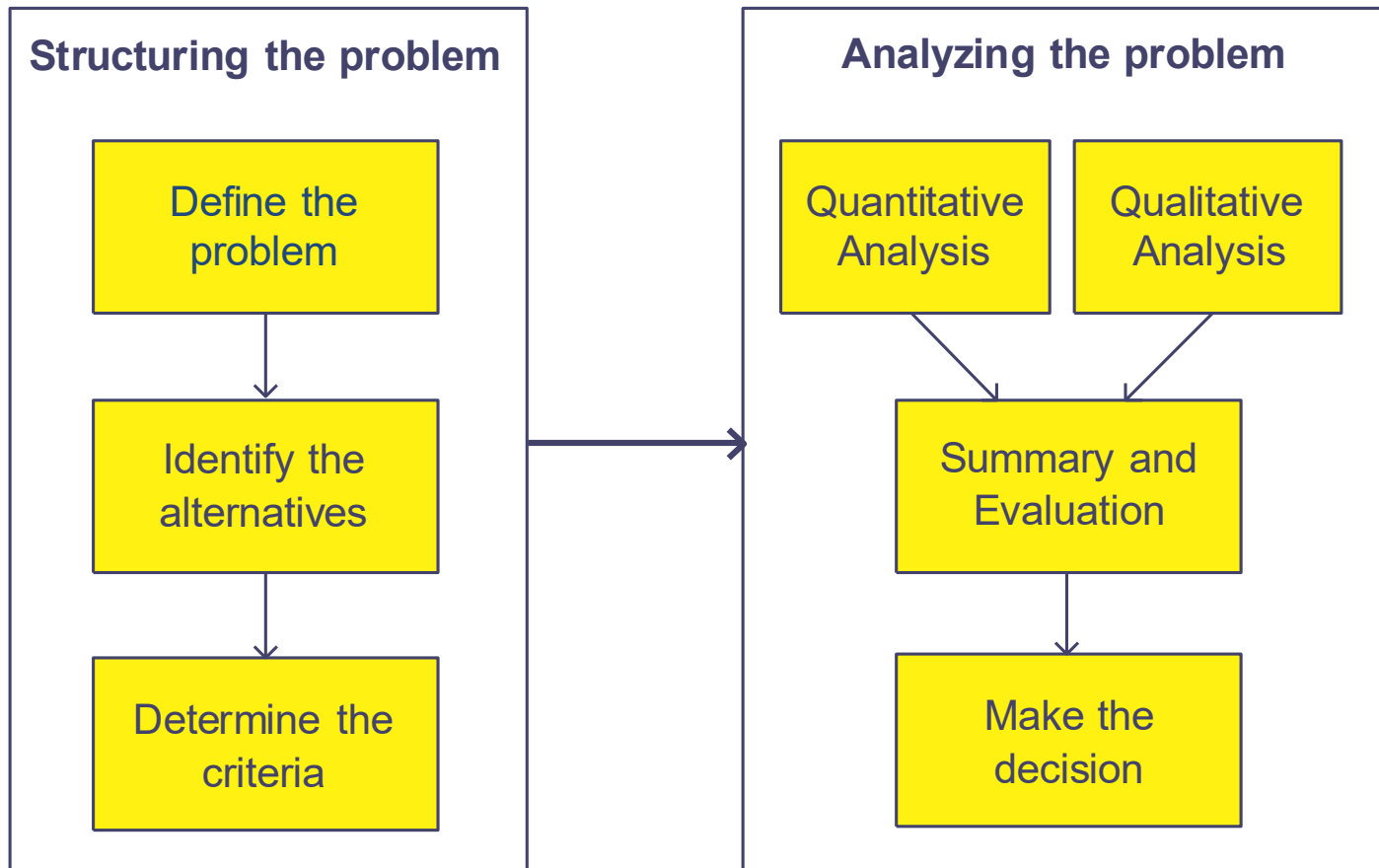
- ❑ Models are constructed to analyze:
 - a system
 - a phenomenon
- ❑ Generally, experimenting with models (compared to experimenting with the real system):
 - requires less time
 - is less expensive
 - involves less risk
- ❑ In general Industrial Engineers work with:
 - production systems (single facility or multiple facility)
 - service systems
 - financial systems

It is only a model!

- A model is never exactly representative! So, its predictions may not be perfect. But models do generate good insights.
- A number of models may be used to see the whole picture...



Decision Making



Quantitative Analysis

□ Quantitative Analysis: procedure

1. model development
2. data preparation
3. model solution
4. analysis of the solution and reporting the results

□ Quantitative Analysis: challenges

1. the decision problem is complex
2. the problem is large scale
3. the decision problem is repetitive

Mathematical Models

- ❑ Quantitative Analysis can be done by using mathematical models
- ❑ **Mathematical models** represent real world problems through a system of mathematical relationships based on:
 - ❑ assumptions,
 - ❑ estimates and
 - ❑ statistical analyses

Mathematical Models

Examples of mathematical models:

- mathematical programming models
- simulation models
- econometric models

Mathematical Programming Models

- ❑ Relate *decision variables* (controllable inputs) with fixed or variable *parameters* (uncontrollable inputs)
- ❑ Maximize or minimize some criteria (*objective function*) subject to limits (*constraints*)
 - **Objectives** (minimize cost, maximize profit, etc.)
 - **Constraints** (capacities, budget limits, etc.)

Guidelines for Mathematical Programming Model Development

- ❑ Write a verbal description of the objective
- ❑ Write a verbal description of each constraints
- ❑ Define the decision variables
- ❑ Express the objective in terms of the decision variables
- ❑ Express the constraints in terms of the decision variables

Modeling is not only mathematics but also an art!

- ❑ First: learning operations research (OR) techniques to develop models and analyze them
- ❑ Next: modeling real systems and using the OR techniques to analyze the models and generate insights for the real systems

Model Components

Model	Decision Variables	Uncontrollable Variables	Outputs
Investment	- Investment quantities - Investment schedules	- Rate of return - Inflation rate - Risk level	- Total return - Total profit - Total risk
Inventory	- When to order - What quantity to order	- Demand - Storage cost - Ordering cost	- Total profit - Service level - Total cost
Transportation	- Quantities shipped - Shipping routes	- Distances - Transportation cost - Demand	Total transportation cost

Example: Automotive Manufacturing Plant

Gather the facts

State them from a systems point of view

- Company manufactures cars and trucks

→ Two types of products

- Each vehicle must be processed in the paint shop and assembly shop

→ There are two processes for all products

Example: Automotive Manufacturing Plant

- It takes 6 minutes to paint a truck, and 10 minutes to paint a car.
- It takes 8 minutes to assemble a truck, and 8 minutes to assemble a car.



→ Processing time of each vehicle

- The factory manufactures trucks and cars for 8 hours a day

→ Available time

Example: Automotive Manufacturing Plant

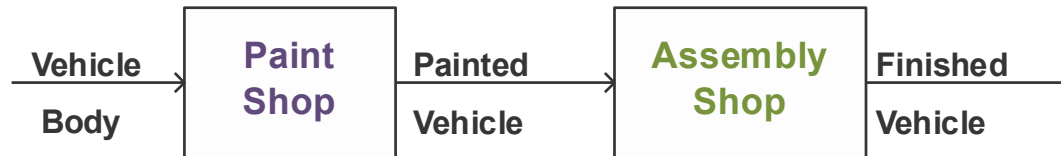
- The marketing division reports that it is impossible to sell more than 40 trucks a day

→ Limitation

- Each truck contributes 3000 TL to profit
- Each car contributes 2000 TL to profit

Example: Process Diagram and Data

□ Process Flow Diagram:



□ Problem Data:

Shop	Processing Time (min/unit)		Available Time (min/day)
	Truck	Car	
Paint	6	10	480
Assembly	8	8	480
Profit (TL/unit)	3000	2000	

Note: The marketing division reports that it would be impossible to sell more than 40 trucks a day.

Example: Model Development

Remember the guidelines for model development:

1. Write a verbal description of the objective
2. Write a verbal description of each constraints
3. Define the decision variables
4. Express the objective in terms of the decision variables
5. Express the constraints in terms of the decision variables

Example: Objective

Shop	Processing Time (min/unit)		Available Time
	Truck	Car	
Paint	6	10	480
Assembly	8	8	480
Profit (TL/unit)	3000	2000	

- ❑ What could be a proper objective for this problem?
- ❑ Review the facts:
 - the factory manufactures two types of vehicles,
 - the manufacturing is composed of two stages: painting, assembly,
 - the factory exists (survives) by making profit.
- ❑ **Objective:** *maximize the total daily profit.*

Example: Constraints

Shop	Processing Time (min/unit)		Available Time
	Truck	Car	
Paint	6	10	480
Assembly	8	8	480
Profit (TL/unit)	3000	2000	

□ Paint Shop:

- both trucks and cars must go through the paint shop,
- the paint shop operates for 8 hours (480 minutes) each day.

□ **Constraint 1:** *Paint shop can operate at most for 480 minutes every day.*

□ Assembly Shop:

- both trucks and cars must go through the assembly shop,
- the assembly shop operates for 8 hours (480 minutes) each day.

□ **Constraint 2:** *Assembly shop can operate at most for 480 minutes every day.*

□ **Constraint 3:** *it is not possible to sell more than 40 trucks a day.*

Example: Decision Variables

□ Decision variables will be used to:

1. express the objective function
2. express the constraints

□ Remark:

- objective function (*maximize the total daily profit*)
is related to the number of trucks and cars produced
- producing a truck and/or a car consumes time available in paint and assembly shops

□ Decision Variables:

- x_1 : number of trucks produced per day [unit/day]
- x_2 : number of cars produced per day [unit/day]

Example: Objective Function Formulation

Shop	Processing Time (min/unit)		Available Time
	Truck	Car	
Paint	6	10	480
Assembly	8	8	480
Profit (TL/unit)	3000	2000	

- **Objective** : *maximize the total daily profit.*
- **Objective** = *profit generated by producing trucks*
+ *profit generated by producing cars*

- profit generated by producing trucks [TL/day]:

$$\begin{aligned} & \text{profit per truck } \left[\frac{TL}{unit} \right] \times \text{no of trucks produced } \left[\frac{unit}{day} \right] \\ & = 3000 \left[\frac{TL}{unit} \right] \times x_1 \left[\frac{unit}{day} \right] \Rightarrow \left[\frac{TL}{day} \right] \end{aligned}$$

- profit generated by producing cars [TL/day]:

$$\begin{aligned} & \text{profit per car } \left[\frac{TL}{unit} \right] \times \text{no of cars produced } \left[\frac{unit}{day} \right] \\ & = 2000 \left[\frac{TL}{unit} \right] \times x_2 \left[\frac{unit}{day} \right] \Rightarrow \left[\frac{TL}{day} \right] \end{aligned}$$

- **Objective** : *maximize* $3000 \times x_1 + 2000 \times x_2 \left[\frac{TL}{day} \right]$

Example: Constraint Formulation

- **Constraint 1** : Paint shop can operate at most for 480 minutes every day.

- **Constraint 1** = *time spent to paint trucks*
+ *time spent to paint cars*
≤ *total available time for painting*

- time spent to paint trucks [min/day]:

$$\begin{aligned} & \text{time per truck } \left[\frac{\text{min}}{\text{unit}} \right] \times \text{no of trucks produced } \left[\frac{\text{unit}}{\text{day}} \right] \\ &= 6 \left[\frac{\text{min}}{\text{unit}} \right] \times x_1 \left[\frac{\text{unit}}{\text{day}} \right] \Rightarrow \left[\frac{\text{min}}{\text{day}} \right] \end{aligned}$$

- time spent to paint cars [min/day]:

$$\begin{aligned} & \text{time per car } \left[\frac{\text{min}}{\text{unit}} \right] \times \text{no of cars produced } \left[\frac{\text{unit}}{\text{day}} \right] \\ &= 10 \left[\frac{\text{min}}{\text{unit}} \right] \times x_2 \left[\frac{\text{unit}}{\text{day}} \right] \Rightarrow \left[\frac{\text{min}}{\text{day}} \right] \end{aligned}$$

- **Constraint 1** : $6 \times x_1 + 10 \times x_2 \leq 480$ [min/day]

Example: Constraint Formulation

- **Constraint 2** : Assembly shop can operate at most for 480 minutes every day.

- **Constraint 2** = *time spent to assemble trucks*
 + *time spent to assemble cars*
 ≤ *total available time for assembling*

- time spent to assemble trucks [min/day]:

$$\begin{aligned} & \text{time per truck } \left[\frac{\text{min}}{\text{unit}} \right] \times \text{no of trucks produced } \left[\frac{\text{unit}}{\text{day}} \right] \\ &= 8 \left[\frac{\text{min}}{\text{unit}} \right] \times x_1 \left[\frac{\text{unit}}{\text{day}} \right] \Rightarrow \left[\frac{\text{min}}{\text{day}} \right] \end{aligned}$$

- time spent to assemble cars [min/day]:

$$\begin{aligned} & \text{time per car } \left[\frac{\text{min}}{\text{unit}} \right] \times \text{no of cars produced } \left[\frac{\text{unit}}{\text{day}} \right] \\ &= 8 \left[\frac{\text{min}}{\text{unit}} \right] \times x_2 \left[\frac{\text{unit}}{\text{day}} \right] \Rightarrow \left[\frac{\text{min}}{\text{day}} \right] \end{aligned}$$

- **Constraint 2** : $8 \times x_1 + 8 \times x_2 \leq 480 \left[\frac{\text{min}}{\text{day}} \right]$

Example: Constraint Formulation

- **Constraint 3** : *it is not possible to sell more than 40 trucks a day.*
 - *the number of trucks produced $\left[\frac{\text{units}}{\text{day}}\right] \leq 40 \left[\frac{\text{units}}{\text{day}}\right]$*
- **Constraint 3** : $x_1 \leq 40 \left[\frac{\text{units}}{\text{day}}\right]$
- Is it possible to have negative production of trucks or cars?

We have an additional consideration on decision variables:

$$x_1 \geq 0 \text{ and } x_2 \geq 0$$

Example: Complete Model

- The **complete mathematical model** for the Automotive Manufacturing Plant:

$$\text{maximize } 3000 \times x_1 + 2000 \times x_2$$

subject to

$$6 \times x_1 + 10 \times x_2 \leq 480$$

$$8 \times x_1 + 8 \times x_2 \leq 480$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Class Exercise: Filsiz Boya Company

- Filsiz Boya company owns a small paint factory that produces both interior and exterior house paints for wholesale distribution.
- Two basic raw materials, R and S, are used to manufacture the paints.
- One ton of exterior paint requires 1 ton of raw material R and 3 tons of material S, whereas 1 ton of interior paint requires 2 tons of raw material R and 1 ton of raw material S.
- The wholesale price per ton is 3000 TL for exterior paint and 2000 TL for interior paint.
- The total amount of raw material R (S) is 6 (11).
- Also, we know that the maximum demand for exterior is 3 tons, whereas the demand for exterior cannot exceed that of interior by more than 1 ton.

Class Exercise: Filsiz Boya Company

- Develop a Linear programming model for Filsiz Boya Company:
 - What will be the **decision variables**?
 - What will be an appropriate **objective function**?
 - What will be the **constraints** to define this system?

Class Exercise: Filsiz Boya Company

- Let X_E and X_I denote the amount of exterior and interior paint to be manufactured by the company (in tons).
- The company aims to maximize the gross income subject to the following constraints:
 - Maximum availability of raw material R (S) is 6 (11).
 - Maximum demand for exterior 3 tons.
 - The demand for exterior cannot exceed that of interior by more than 1 ton.

Filsiz Boya: Summary of the Info

- *Raw materials*

- $X_E \rightarrow 1 \text{ R \& 3 S}$ S is at most 11 tons
- $X_I \rightarrow 2 \text{ R \& 1 S}$ R is at most 6 tons

- *Prices:*

- $X_E \rightarrow 3000 \text{ TL}$
- $X_I \rightarrow 2000 \text{ TL}$

- *Demand*

- Demand for exterior is at most 3 tons
- Demand for exterior is at most (demand for interior + 1) tons

Filsiz Boya: Objective

- To maximize the gross income
- Price per ton:
 - Exterior paint: 3000 TL
 - Interior paint: 2000 TL
- X_E (X_I) : amount of exterior (interior) paint to be manufactured (in tons)

$$\text{maximize } 3000 \times x_E + 2000 \times x_I$$

Filsiz Boya: Constraints

- Maximum availability of raw material R (S) is 6 (11).
- Maximum demand for exterior 3 tons.
- The demand for exterior cannot exceed that of interior by more than 1 ton.

- Raw material:

$$x_E \rightarrow 1 \text{ R \& 3 S}$$

$$x_I \rightarrow 2 \text{ R \& 1 S}$$

$$1 \times x_E + 2 \times x_I \leq 6$$

$$3 \times x_E + 1 \times x_I \leq 11$$

- Demand:

$$x_E \leq 3$$

$$x_E \leq x_I + 1$$

- Non-negativity:

$$x_E \geq 0, \quad x_I \geq 0$$

Filsiz Boya: Complete Model

$$\text{maximize } 3000 \times x_E + 2000 \times x_I$$

subject to

$$\begin{array}{rcl} x_E + & 2 x_I & \leq 6 \\ 3 x_E + & x_I & \leq 11 \\ x_E & & \leq 3 \\ x_E - & x_I & \leq 1 \\ -x_E & & \leq 0 \\ & -x_I & \leq 0 \end{array}$$

QUESTIONS AND COMMENTS